INTERNATIONAL DEVELOPMENT ASSOCIATION
INTERNATIONAL MONETARY FUND

KENYA

Joint IMF/World Bank Debt Sustainability Analysis

Prepared by the Staffs of the International Monetary Fund and
the World Bank

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This analysis updates the previous joint Bank/Fund debt sustainability analysis (DSA) submitted to the Executive Board in January 2011(Country Report No. 11/48).[1] Compared to the January 2011 analysis, overall debt sustainability indicators have not changed significantly. Lower than anticipated fiscal deficit in 2010 was offset by an increase in external debt-to- GDP ratio due to exchange rate depreciation and increased IMF financing. Risks are somewhat greater for external debt, particularly in the historical scenario, as the current account deficit has widened in 2010. Nevertheless, Kenya remains at low risk of external and domestic debt distress.[2,3] Strategies to guard against shocks could include a build-up in international reserves as envisaged in the ECF framework.

I. BACKGROUND

1. **At end-2010, nominal public external debt was estimated at $7.3 billion (24 ¼ percent of GDP)**. About 60 percent of this debt was to multilateral creditors (including 43 percent owed to the World Bank) and 39 percent to bilateral creditors. A small share of this debt (under 2 percent) represents disputed arrears on security-related contracts, owed to commercial creditors.

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[1] It has benefited from consultation with African Development bank staff.

[2] World Bank classifies Kenya as a medium performer in terms of the quality of its policies and institutions as measured by a three-year average of the World Bank's Country Policy and Institutional Assessment (CPIA) index. (available at http://go.worldbank.org/AXO6I14PK0)

[3] For a medium performer, the indicative thresholds for external debt sustainability are a net present value (NPV) of debt-to-GDP ratio of 40 percent, an NPV of debt-to-exports ratio of 150 percent, an NPV of debt-to-revenue ratio of 250 percent, a debt service-to-exports ratio of 20 percent, and a debt service-to-revenue ratio of 30 percent.

2. **Kenya has managed its debt relatively well and has regularly met its obligations, except for some disputed commercial arrears**. Its external debt-to GDP ratio was 24 percent at end-2010, mainly a result of prudent external borrowing, and Kenya has a more manageable debt ratio than most of its low-income country peers. Kenya did not qualify for heavily indebted poor countries (HIPC) debt relief, as its debt indicators have been below the HIPC Initiative thresholds, but it benefited from Paris club rescheduling in 2004.

3. **The disputed external commercial arrears, estimated at US$242 million, are subject of ongoing investigations and litigation**. The timeline for clearance of these arrears has not been determined. The amount of arrears has been revised upward from an earlier estimate of US$91 million, following completion of independent valuation of works, goods, and services delivered under each contract. The authorities think that it is more prudent to estimate a higher figure to reflect the likelihood of court rulings in favor of all creditors.

4. **Kenya's net domestic debt stood at Ksh 580 billion at end-2010 (24¾ percent of GDP), but potential domestic contingent liabilities due to pensions could be large.** During 2003–07, domestic debt declined to 13½ percent of GDP, as a result of strong economic growth, prudent fiscal policies, and lower interest rates. The downward trend was reversed during 2008–09 reflecting fiscal stimulus measures implemented to mitigate the impact of the global financial crisis. Despite relatively low domestic debt-to-GDP ratio, Kenya may be vulnerable to possible realization of contingent liabilities associated with unfunded obligations of the National Social Security Fund and government's current pay-as-you-go pension scheme for civil servants as well as domestic debt of parastatal companies.

5. **The DSA is based on non-reconciled debt data provided by the authorities, and staff estimates**. It consists of two parts**—external and fiscal**.

- The external DSA covers external debt of the central government and the central bank. External debt sustainability is assessed in relation to policy-dependent debt-burden thresholds. A single discount rate of 4 percent is used.

- The fiscal DSA covers total debt—external and domestic—incurred or guaranteed by the central government.[4]

[4] Public domestic debt comprises central government debt. In this analysis, total public debt refers to the sum of public domestic and public external debt, but does not cover the entire public sector (e.g., parastatal borrowing without a government guarantee is not covered).

II. MACROECONOMIC ASSUMPTIONS

6. **Key medium-term assumptions underlying the DSA are consistent with the Staff Report for the 2011 Article IV Consultations, Second Review Under the Three-Year Arrangement Under the Extended Credit Facility and Request for Modification of Performance Criteria and Augmentation of Access. These assumptions are:**

- Real GDP growth is projected at around 5 percent for 2011, 5 ½ in 2012, and about 6 percent afterwards. The projected growth remains, on average, about the same as the pre-crisis five-year average of just above 6 percent and represents a deceleration from the 2007 growth of 7 percent. As such, the projected growth reflects in part a resumption of the momentum that was abruptly disrupted by the 2007 post-elections violence.[5]

- Average inflation, as measured by the GDP deflator, is projected at about 12 percent in 2012, 7 percent for 2013, and 5 percent after 2014.

- The exchange rate is assumed to remain constant in real terms over the medium term.

- The noninterest external current account deficit rises to about 9¾ percent of GDP in 2011, before falling to about 3 percent of GDP by 2015.

- Broadly constant revenue and grants as a share of GDP (about 26 percent);[6] the primary fiscal deficit is 1.5 percent of GDP in 2010/11 and gradually declines to 1.1 percent by the end of the forecast period.

- Development spending increasing from just about 8.0 percent of GDP in 2010 to about 10 percent by 2012; a constant wage bill of 7.1 percent of GDP after 2012 compared to 6.8 percent in the 2011/12 budget, and a gradual decline in other recurrent spending from 8.9 percent of GDP in 2008 to 7.2 percent of GDP over the long-term, in line with the government's budget strategy.

- Real interest rates on domestic public debt are assumed at about 3 percent for short-term debt and about 5 percent for medium- and long-term debt.

- New domestic borrowings consists of 30 percent short-term debt and 70 percent of medium- and long-term debt, with the latter having an average maturity of about seven years. The NPV of domestic debt is assumed to be equal to its face value.

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[5] The likelihood of the domestic shocks of this nature has been reduced following the August 2010 ratification of the new constitution.

[6] The increase from the average of about 23 percent of GDP in the past three years reflect an expected improvement in the revenue mobilization effort stemming from tax reforms, as well as in the capacity to absorb project grants.

- New external borrowing as a share of GDP (including nonconcessional borrowing described below) increases over the medium and then declines gradually. It is projected to average 3 percent of GDP during 2011–15, up from 2 ¼ percent of GDP in 2010. It subsequently declines, falling below 1 percent by the end of the forecast period. New external borrowing assumptions include sovereign bond issuance of $500 million in 2012, additional commercial borrowing of about US$450 million during 2013–14 and about US$200 million per year in the long run. Assumptions on the debt terms include a 7½ percent fixed interest rate 10 year maturity and a bullet amortization.

7. Continued eligibility for concessional borrowing from the International Development Association is assumed, although achievement of assumed growth rates could imply graduation during the forecast period.

III. External Debt Sustainability

8. **Kenya faces a low risk of external debt distress reflecting the limited reliance on external borrowing and an expected improvement in macroeconomic performance**.

- Under the *baseline scenario***,** initial debt ratios are well below all of the indicative thresholds for a medium performer, even if they increase over the medium-term reflecting a higher rate of debt accumulation (see Figure 1 and Table 2a and 2b).

- **Alternative scenarios and stress tests indicate that Kenya's external debt situation is generally resilient.** Standard stress tests reveal an initial upward trend for the debt indicators but do not result in a breach of the thresholds during the projection period. Over the period 2011–15, a shock combining lower GDP growth, weaker exports, a lower GDP deflator, and a fall in non-debt creating flows would push the NPV of public external debt as a share of GDP from 18 percent to 26 percent, and the NPV of debt-to-exports from almost 57 percent to 89 percent. The most extreme shocks to debt dynamics by 2021 generally stem from a one-time 30 percent depreciation in 2011 or from a one standard deviation shock to the growth of exports proceeds.[7]

[7] The most extreme shock to the NPV of debt-to-GDP, the NPV of debt-to-revenue, and debt service-to-revenue results from a 30 percent exchange rate depreciation in 2011, whereas the most extreme shock to the PV of debt-to-exports results from an exports growth subdued during 2011–12 at only 1.3 percent (the historical average minus one standard deviation).

Summary: External Debt Sustainability Assessment
(Percent of GDP)

	2011	2012	2013	2014	2015	2016
NPV of PPG External Debt						
In percent of GDP (threshold=40)						
Baseline	17.7	18.7	19.1	18.9	18.7	18.0
Combined shocks	17.7	21.8	26.1	25.5	25.0	23.9
In percent of exports (threshold=150)						
Baseline	57.0	65.3	68.9	73.4	76.4	78.2
Combined shocks	57.0	69.1	81.0	85.3	87.9	89.5
PPG External Debt Service						
In percent of exports (threshold=20)						
Baseline	3.8	4.2	4.2	4.7	4.8	4.9
Combined shocks	3.8	4.1	4.3	4.9	5.0	5.2

IV. PUBLIC DEBT SUSTAINABILITY

9. **Kenya's public debt shows some vulnerability to growth shocks.**

- Under the *baseline scenario*, the NPV of total public debt-to-GDP, at 40½ percent in 2010, increases and peaks at almost 42 percent in 2012 and gradually trends down to 40 percent of GDP by 2014. Afterwards, it trends down to around 25 percent by 2031 (Figure 2 and Table 1a). Given Kenya's relatively strong revenue performance, the NPV of debt-to-revenue ratio declines to below 150 percent in 2013. The debt service-to-revenue ratio falls to 22 percent by 2015, from 25 in 2009. It declines to below 20 percent by 2030.

- *Alternative scenarios* and *stress tests* indicate that Kenya's debt indicators are vulnerable to slower growth, keeping the primary balance fixed to a historical average, and materialization of contingent liabilities (see Figure 2 and Table 1b). A scenario assuming that 10 percent of 2010 GDP in potential domestic currency liabilities as of end-2010 would be paid by the government in equal tranches over a 10-year period shows that debt indicators deteriorate notably compared with the baseline. An alternative scenario shows that a two-year growth shock leads to a rise in the NPV of debt-to-GDP ratio to 55 percent by 2014, an NPV of debt-to-revenue ratio to over 200 percent by 2020, and a rise in the debt service-to-revenue ratio to over 31 percent by 2030. Also, the scenario of permanently lower growth—baseline minus half a percentage point—results in debt indicators that are considerably higher in the long-term (e.g., by the end of the forecast period, the NPV of debt-to-GDP ratio would be 50 percent). This result reinforces the importance of implementing fiscal

consolidation and expanding productive capacity in the medium term, in addition to pursuing a prudent borrowing approach, to avoid a rising debt burden.

V. CONCLUSIONS

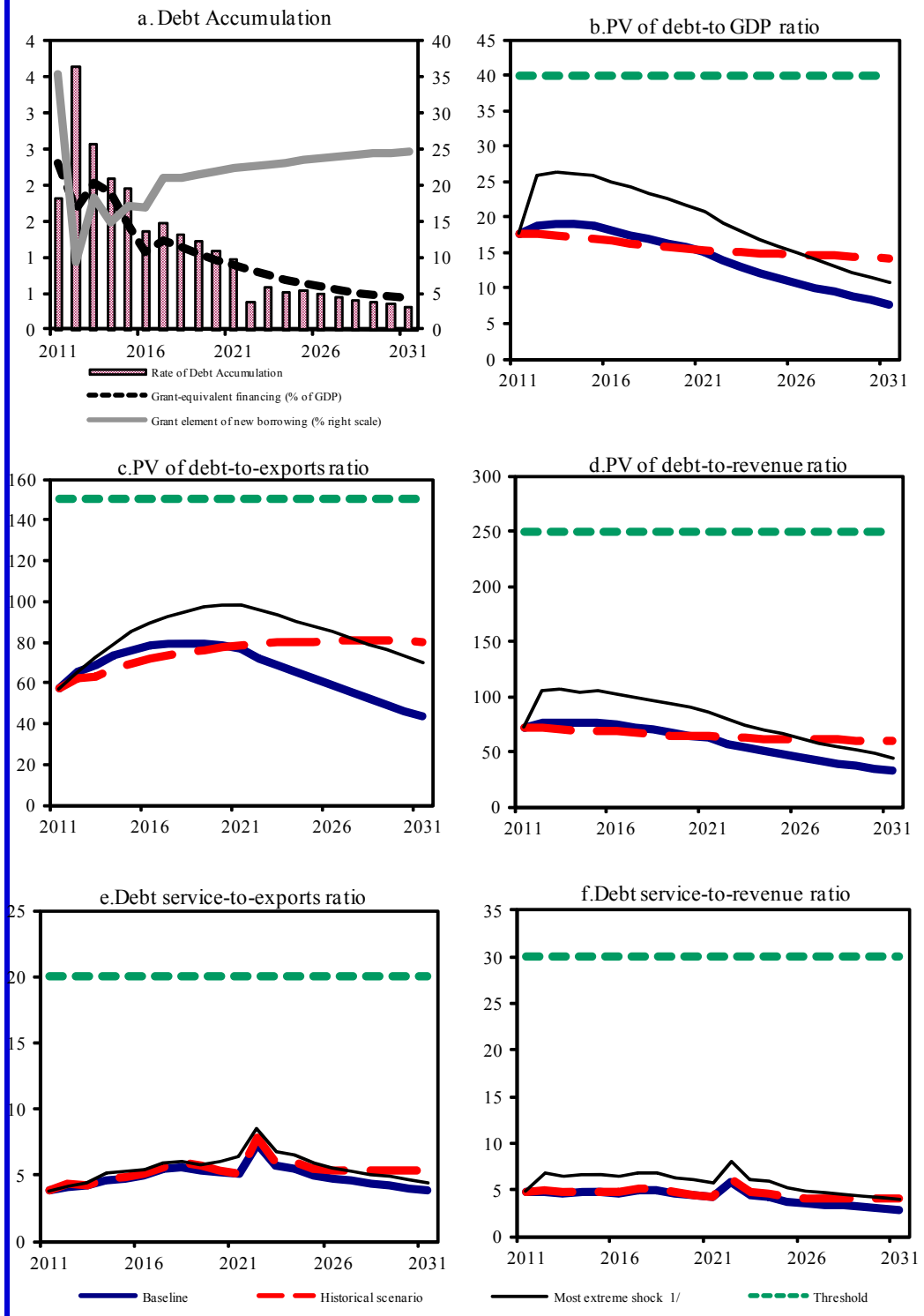
10. **Kenya faces a low risk of external debt distress, reflecting the limited reliance on external borrowing and an expected improvement in macroeconomic performance**. All external public debt indicators remain below the relevant country-specific debt burden thresholds. Further, although standard stress tests reveal a worsening in debt indicators, they do not result in a breach of the thresholds during the projection period.

11. **Total public debt, however shows greater risk of unfavorable debt developments, especially under a shock to GDP growth, unchanged fiscal policy, or materialization of some contingent liabilities**. Even temporarily lower GDP growth would set the NPV of public debt-to-GDP, the NPV of debt-to-revenue, and the ratio of debt service-to-revenue on a sharply increasing trend. A permanently unchanged primary balance from its 2010 level worsens debt dynamics notably. Potentially large but unreported contingent liabilities also pose additional risks to the sustainability of public debt.

12. **The sustainability of Kenya's debt depends on macroeconomic performance and a prudent borrowing strategy**. The projected investment in infrastructure and the assumed improvement in the investment climate would be crucial to sustaining strong exports and GDP growth. Additionally, Kenya's success in avoiding unsustainable debt to date reflects good management, but also limited willingness on the part of creditors to provide financing, at times due to governance concerns.

13. **The authorities were involved in the DSA exercise and concur with its conclusions.** The staffs encourage Kenyan authorities to build on their recent medium-term debt strategy and to use tools such as the joint IMF/WB DSA template to help maintain a prudent borrowing strategy. Such a strategy should continue to consider the total concessionality and interest costs of Kenya's borrowing, maturity structure, and steps that would help guard against volatility, whether due to shocks such as droughts or to fluctuations in external assistance. Strategies to guard against shocks could include a build-up in international reserves as envisaged in the ECF framework.

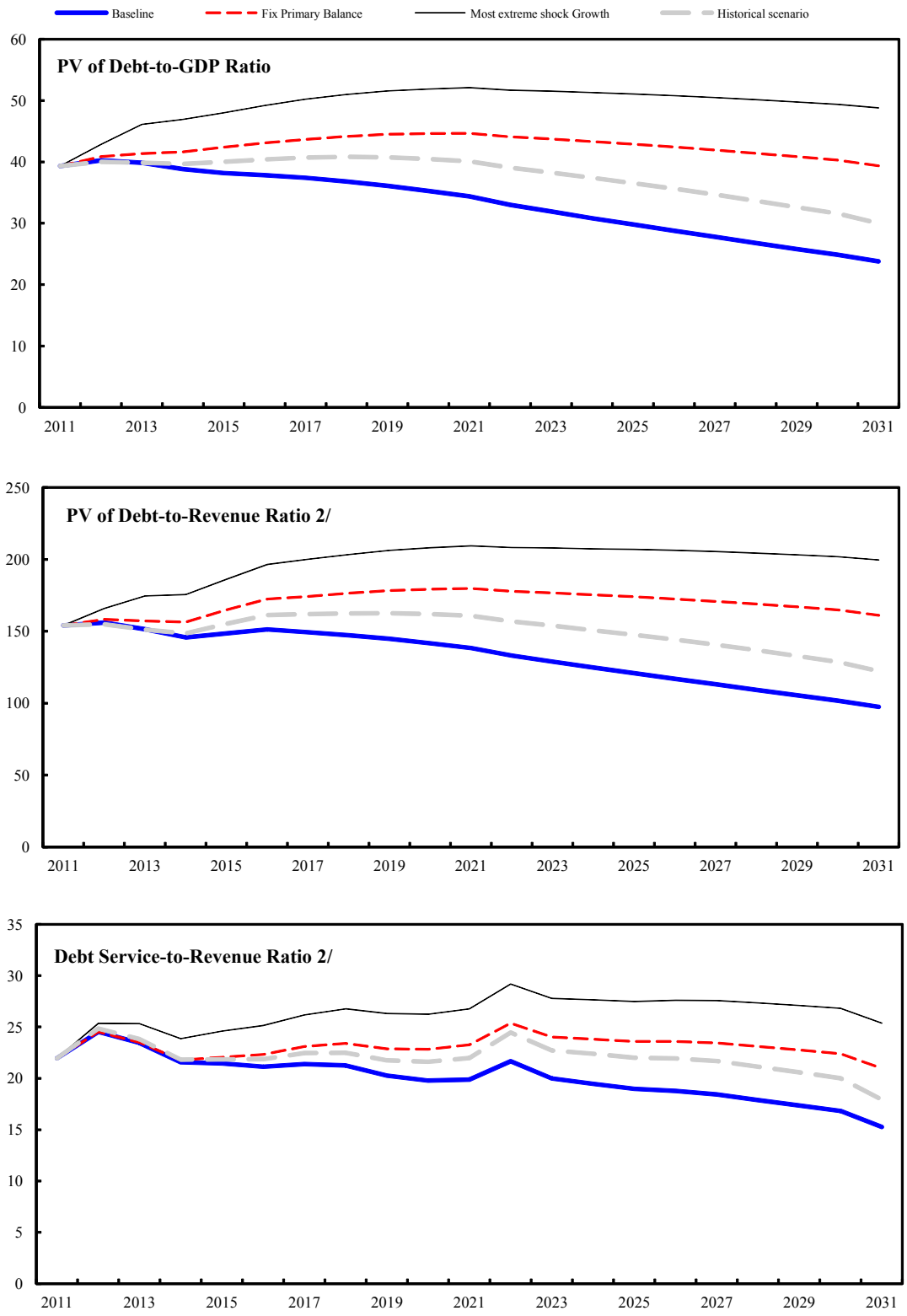
Figure 1. Kenya: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2011–31[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2021. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to a Non-debt flows shock and in figure f. to a One-time depreciation shock

Figure 2.Kenya: Indicators of Public Debt Under Alternative Scenarios, 2011-2031 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2021.
2/ Revenues are defined inclusive of grants.

Table 1a.Kenya: Public Sector Debt Sustainability Framework, Baseline Scenario, 2008–31
Percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2008	2009	2010	Average[5]	Standard Deviation[5]	2011	2012	2013	2014	2015	2016	2011–16 Average	2021	2031	2017–31 Average
Public sector debt[1]	41.3	42.7	45.7			47.5	47.5	46.7	45.0	43.8	43.1		39.0	28.1	
Off Which : foreign-currency denominated	23.7	22.9	22.9			25.9	25.9	25.9	25.2	24.6	23.5		19.3	10.2	
Change in public sector debt	2.1	1.4	3.0			1.8	0.0	-0.8	-1.8	-1.2	-0.7		-0.9	-1.0	
Identified debt-creating flows	2.1	-0.6	3.2			1.0	-2.2	-1.2	-1.8	-1.7	-1.2		-0.7	0.5	
Primary deficit	2.1	3.1	2.8	1.0	1.9	2.5	2.3	1.9	1.2	1.0	1.5	1.8	1.4	1.9	1.6
Revenue and grants	23.2	22.8	24.6			25.5	25.8	26.3	26.6	25.7	25.0		24.7	24.1	
Of which: grants	1.1	0.9	0.8			1.0	1.3	1.4	1.5	1.0	0.6		0.5	0.2	
Primary (noninterest) expenditure	25.4	25.8	27.4			28.0	28.1	28.2	27.8	26.7	26.4		26.1	26.1	
Automatic debt dynamics	1.7	-3.7	0.3			-1.5	-4.6	-3.1	-3.0	-2.8	-2.7		-2.1	-1.5	
Contribution from interest rate/growth differential	-1.2	-1.4	-0.7			-3.2	-3.0	-2.1	-2.0	-1.9	-2.0		-1.5	-1.1	
Of which: contribution from average real interest rate	-0.6	-0.3	1.5			-1.0	-0.5	0.6	0.8	0.8	0.7		0.7	0.5	
Of which: contribution from real GDP growth	-0.6	-1.1	-2.2			-2.2	-2.5	-2.7	-2.8	-2.7	-2.7		-2.3	-1.7	
Contribution from real exchange rate depreciation	2.9	-2.3	1.0			1.7	-1.6	-1.0	-1.0	-0.8	-0.7		...	...	
Other identified debt-creating flows	-1.8	0.1	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-1.8	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.1	0.1	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.0	2.0	-0.2			0.8	2.3	0.4	0.0	0.5	0.5		-0.2	-1.5	
Other Sustainability Indicators															
PV of public sector debt	...	...	38.5			39.3	40.3	39.9	38.7	37.9	37.6		34.7	25.6	
Off which : foreign-currency denominated	...	...	15.8			17.7	18.7	19.1	18.9	18.7	18.0		15.0	7.7	
Off which : external	...	...	15.8			17.7	18.7	19.1	18.9	18.7	18.0		15.0	7.7	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need[2]	11.4	17.6	14.8			15.2	15.4	15.1	13.8	13.2	13.2		13.0	11.9	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	156.5			154.2	156.1	151.5	145.6	147.8	150.5		140.4	106.1	
PV of public sector debt-to-revenue ratio (in percent)	…	…	161.6			160.5	164.3	160.0	154.2	153.6	154.5		143.3	107.1	
Off which : external[3]	…	…	66.1			72.2	76.2	76.6	75.4	75.8	74.0		62.0	32.3	
Debt service-to-revenue and grants ratio (in percent)[4]	18.4	41.7	25.5			21.9	24.5	23.5	21.6	21.5	21.1		20.0	15.5	
Debt service-to-revenue ratio (in percent)[4]	19.4	43.3	26.4			22.8	25.8	24.8	22.9	22.3	21.7		20.4	15.6	
Primary deficit that stabilizes the debt-to-GDP ratio	0.1	1.7	-0.1			0.7	2.3	2.7	3.0	2.2	2.2		2.4	3.0	
Key macroeconomic and fiscal assumptions														1.1	
Real GDP growth (in percent)	1.5	2.6	5.6	4.2	2.2	5.0	5.6	6.0	6.4	6.4	6.5	6.0	6.1	6.1	6.1
Average nominal interest rate on forex debt (in percent)	1.5	0.5	1.0	1.4	1.0	1.0	1.3	1.5	1.7	1.9	2.0	1.6	2.4	2.1	2.2
Average real interest rate on domestic debt (in percent)	-1.7	0.0	8.4	2.2	3.1	-3.6	-2.5	1.8	3.2	3.3	3.3	0.9	3.1	2.7	3.1
Real exchange rate depreciation (in percent, + indicates depreciation)	14.1	-10.3	4.8	-2.5	7.9	7.9	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	11.9	10.9	2.2	5.9	3.7	12.1	12.3	7.1	5.6	5.2	5.0	7.9	5.0	5.0	5.0
Growth of real primary spending (deflated by GDP deflator, in percent)	0.1	0.0	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.1	0.1	0.1	0.1	0.1
Grant element of new external borrowing (in percent)	...	...	...	…	…	35.4	9.3	18.3	14.9	17.2	17.0	18.7	22.3	24.7	...

Sources: Country authorities; and staff estimates and projections.

[1] Public debt refers to net debt of the central government.

[2] Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

[3] Revenues excluding grants.

[4] Debt service is defined as the sum of interest and amortization of medium and long-term debt.

[5] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 1b.Kenya: Sensitivity Analysis for Key Indicators of Public Debt 2011–31

	Projections							
	2011	2012	2013	2014	2015	2016	2021	2031
PV of Debt-to-GDP Ratio								
Baseline	39	40	40	39	38	38	35	26
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	39	40	40	40	40	40	40	30
A2. Primary balance is unchanged from 2011	39	41	41	42	42	43	45	39
A3. Permanently lower GDP growth[1]	39	41	41	41	41	41	44	47
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012–13	39	43	46	47	48	49	52	50
B2. Primary balance is at historical average minus one standard deviations in 2012–13	39	41	42	41	41	40	38	29
B3. Combination of B1-B2 using one half standard deviation shocks	39	41	42	43	43	44	46	43
B4. One-time 30 percent real depreciation in 2012	39	48	47	46	45	44	41	31
B5. 10 percent of GDP increase in other debt-creating flows in 2012	39	50	49	48	47	46	42	32
PV of Debt-to-Revenue Ratio[2]								
Baseline	154	156	151	146	148	151	140	106
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	154	155	151	149	156	161	162	122
A2. Primary balance is unchanged from 2011	154	158	157	157	165	173	181	162
A3. Permanently lower GDP growth1	154	159	157	154	159	166	176	195
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012–13	154	166	175	175	185	196	212	209
B2. Primary balance is at historical average minus one standard deviations in 2012–13	154	160	160	155	158	162	154	121
B3. Combination of B1-B2 using one half standard deviation shocks	154	160	160	160	168	177	188	179
B4. One-time 30 percent real depreciation in 2012	154	185	179	172	174	177	165	130
B5. 10 percent of GDP increase in other debt-creating flows in 2012	154	193	187	180	183	186	171	131
Debt Service-to-Revenue Ratio[2]/								
Baseline	22	25	23	22	21	21	20	15
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	22	25	24	22	22	22	22	17
A2. Primary balance is unchanged from 2011	22	25	24	22	22	22	23	20
A3. Permanently lower GDP growth1	22	25	24	22	22	22	23	24
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012–13	22	25	25	24	25	25	27	26
B2. Primary balance is at historical average minus one standard deviations in 2012–13	22	25	24	22	22	22	20	16
B3. Combination of B1-B2 using one half standard deviation shocks	22	25	25	22	23	23	24	22
B4. One-time 30 percent real depreciation in 2012	22	25	25	24	24	24	23	19
B5. 10 percent of GDP increase in other debt-creating flows in 2012	22	25	26	26	26	25	22	17

Sources: Country authorities; and staff estimates and projections.

[1] Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

[2] Revenues are defined inclusive of grants.

Table 2a.: External Debt Sustainability Framework, Baseline Scenario, 2008-2031[1]
(In percent of GDP, unless otherwise indicated)

	Actual			Historical[6] Average	Standard[6] Deviation	Projections									
	2008	2009	2010			2011	2012	2013	2014	2015	2016	2011–16 Average	2021	2031	2017–31 Average
External debt (nominal)[1]/	**23.7**	**22.9**	**22.9**			**25.9**	**25.9**	**25.9**	**25.2**	**24.6**	**23.5**		**19.3**	**10.2**	
Of which : public and publicly guaranteed (PPG)	23.7	22.9	22.9			25.9	25.9	25.9	25.2	24.6	23.5		19.3	10.2	
Change in external debt	2.6	-0.8	0.0			2.9	0.1	0.0	-0.8	-0.6	-1.1		-0.9	-0.7	
Identified net debt-creating flows	2.5	3.6	4.5			7.0	6.6	3.1	3.9	0.4	0.8		0.7	-0.8	
Non-interest current account deficit	**6.4**	**5.3**	**6.5**	**2.5**	**2.9**	**9.9**	**9.5**	**5.9**	**6.8**	**3.2**	**3.6**		**3.1**	**0.6**	2.4
Deficit in balance of goods and services	14.2	12.6	14.5			18.2	17.7	15.9	16.7	13.9	15.3		14.1	9.3	
Exports	27.6	24.2	26.9			31.0	28.6	27.7	25.8	24.5	23.0		19.6	17.7	
Imports	41.8	36.8	41.4			49.2	46.3	43.6	42.5	38.5	38.3		33.7	27.0	
Net current transfers (negative = inflow)	-7.7	-7.3	-7.6	-6.8	0.9	-7.0	-7.4	-8.1	-8.5	-9.0	-8.9		-8.3	-6.5	-7.9
Of which : official	0.1	0.1	0.1			0.1	0.1	0.1	0.0	0.0	0.0		0.0	0.0	
Other current account flows (negative = net inflow)	-0.1	0.0	-0.4			-1.3	-0.9	-1.9	-1.4	-1.7	-2.9		-2.7	-2.2	
Net FDI (negative = inflow)	**-2.2**	**-1.5**	**-1.2**	**-1.4**	**1.0**	**-2.0**	**-1.9**	**-1.9**	**-1.8**	**-1.8**	**-1.7**		**-1.7**	**-1.0**	-1.5
Endogenous debt dynamics[2]	**-1.8**	**-0.2**	**-0.9**			**-0.9**	**-1.0**	**-1.0**	**-1.1**	**-1.0**	**-1.0**		**-0.7**	**-0.4**	
Contribution from nominal interest rate	0.3	0.2	0.3			0.3	0.3	0.4	0.4	0.4	0.4		0.4	0.2	
Contribution from real GDP growth	-0.3	-0.6	-1.2			-1.2	-1.3	-1.4	-1.5	-1.4	-1.4		-1.1	-0.6	
Contribution from price and exchange rate changes	-1.7	0.2	0.1			…	…	…	…	…	…		…	…	
Residual (3-4)[3]	**0.1**	**-4.4**	**-4.4**			**-4.1**	**-6.5**	**-3.1**	**-4.6**	**-1.0**	**-1.9**		**-1.6**	**0.2**	
Of which : exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.4		0.0	0.0	
PV of external debt[4]	...	...	15.8			17.7	18.7	19.1	18.9	18.7	18.0		15.0	7.7	
In percent of exports	...	...	58.5			57.0	65.3	68.9	73.4	76.4	78.2		76.5	43.5	
PV of PPG external debt	**...**	**...**	**15.8**			**17.7**	**18.7**	**19.1**	**18.9**	**18.7**	**18.0**		**15.0**	**7.7**	
In percent of exports	**...**	**...**	**58.5**			**57.0**	**65.3**	**68.9**	**73.4**	**76.4**	**78.2**		**76.5**	**43.5**	
In percent of government revenues	**...**	**...**	**66.1**			**72.2**	**76.2**	**76.6**	**75.4**	**75.8**	**74.0**		**62.0**	**32.3**	
Debt service-to-exports ratio (in percent)	**10.8**	**19.0**	**5.0**			**5.1**	**4.8**	**4.8**	**5.2**	**5.2**	**5.2**		**5.1**	**3.9**	
PPG debt service-to-exports ratio (in percent)	**7.8**	**17.6**	**4.0**			**3.8**	**4.2**	**4.2**	**4.7**	**4.8**	**4.9**		**5.1**	**3.9**	
PPG debt service-to-revenue ratio (in percent)	**9.8**	**19.5**	**4.5**			**4.8**	**4.9**	**4.7**	**4.8**	**4.7**	**4.7**		**4.2**	**2.9**	
Total gross financing need (Billions of U.S. dollars)	2.2	2.6	2.1			3.0	3.2	2.2	2.8	1.3	1.7		2.3	0.7	
Non-interest current account deficit that stabilizes debt ratio	3.9	6.1	6.5			7.0	9.4	5.9	7.5	3.8	4.7		4.0	1.3	
Key macroeconomic assumptions															
Real GDP growth (in percent)	1.5	2.6	5.6	4.2	2.2	5.0	5.6	6.0	6.4	6.4	6.5	6.0	6.1	6.1	6.1
GDP deflator in US dollar terms (change in percent)	8.9	-0.8	-0.4	5.6	5.9	-6.4	8.1	5.0	5.2	4.8	4.6	3.6	4.7	4.6	4.7
Effective interest rate (percent)[5]	1.3	0.8	1.3	1.4	0.9	1.2	1.4	1.6	1.7	1.9	2.0	1.6	2.4	2.1	2.2
Growth of exports of G&S (US dollar terms, in percent)	17.4	-10.6	16.8	12.5	9.4	13.2	5.4	7.8	4.0	6.1	4.7	6.9	8.3	10.6	9.2
Growth of imports of G&S (US dollar terms, in percent)	24.6	-10.4	18.4	14.3	13.4	16.7	7.5	4.8	8.9	1.0	11.2	8.3	8.3	8.8	8.5
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	35.4	9.3	18.3	14.9	17.2	17.0	18.7	22.3	24.7	23.0
Government revenues (excluding grants, in percent of GDP)	22.1	21.9	23.8			24.5	24.5	24.9	25.1	24.7	24.3		24.2	23.9	24.1
Aid flows (in Billions of US dollars)[7]	0.3	0.6	0.2			0.7	0.7	0.8	0.9	0.6	0.4		0.9	1.4	
Of which : Grants	0.3	0.3	0.2			0.3	0.5	0.6	0.7	0.5	0.4		0.5	0.6	
Of which : Concessional loans	0.0	0.4	0.0			0.4	0.2	0.3	0.2	0.1	0.1		0.4	0.8	
Grant-equivalent financing (in percent of GDP)[8]	...	...	...			2.3	1.7	2.0	1.9	1.4	1.1		0.9	0.4	0.7
Grant-equivalent financing (in percent of external financing)[8]	...	...	...			49.0	30.7	41.7	44.2	38.9	33.7		40.1	39.6	39.7
Memorandum items:															
Nominal GDP (Billions of US dollars)	30.0	30.6	32.2			31.6	36.1	40.1	44.9	50.1	55.9		94.8	270.7	
Nominal dollar GDP growth	10.5	1.8	5.2			-1.7	14.1	11.3	11.9	11.5	11.5	9.8	11.0	11.1	11.1
PV of PPG external debt (in Billions of US dollars)			5.0			5.6	6.7	7.7	8.5	9.4	10.1		14.2	20.9	
(PVt-PVt-1)/GDPt-1 (in percent)						1.8	3.6	2.6	2.1	2.0	1.4	2.2	1.0	0.3	0.7
Gross workers' remittances (Billions of US dollars)	1.0	1.0	1.1			1.2	1.3	1.6	1.8	2.0	2.3		3.6	8.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	15.2			17.0	18.0	18.4	18.2	18.0	17.3		14.5	7.5	
PV of PPG external debt (in percent of exports + remittances)	...	...	51.8			50.9	58.1	60.1	63.7	65.6	66.5		64.1	37.3	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	3.6			3.4	3.7	3.7	4.0	4.1	4.2		4.3	3.3	

Sources: Country authorities; and staff estimates and projections.

[1] Includes both public and private sector external debt.

[2] Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

[3] Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

[4] Assumes that PV of private sector debt is equivalent to its face value.

[5] Current-year interest payments divided by previous period debt stock.

[6] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

[7] Defined as grants, concessional loans, and debt relief.

[8] Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2b.Kenya: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011–31
(Percent)

				Projections				
	2011	2012	2013	2014	2015	2016	**2021**	2031
PV of debt-to GDP ratio								
Baseline	17.7	18.7	19.1	18.9	18.7	18.0	**15.0**	7.7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011–31[1]	17.7	17.7	17.4	17.2	16.9	16.6	**15.4**	14.2
A2. New public sector loans on less favorable terms in 2011–31[2]	17.7	18.5	20.0	20.4	20.8	20.6	**19.3**	12.5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012–13	17.7	19.3	20.5	20.3	20.1	19.4	**16.1**	8.3
B2. Export value growth at historical average minus one standard deviation in 2012–13[3]	17.7	19.2	21.1	20.7	20.4	19.6	**16.0**	7.9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012–13[3]	17.7	20.3	21.8	21.6	21.4	20.6	**17.1**	8.8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012–13[4]	17.7	21.2	24.5	23.9	23.3	22.3	**17.8**	8.3
B5. Combination of B1-B4 using one-half standard deviation shocks	17.7	21.8	26.1	25.5	25.0	23.9	**19.3**	9.2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012[5]	17.7	25.9	26.4	26.2	25.9	24.9	**20.8**	10.7
PV of debt-to-exports ratio								
Baseline	57.0	65.3	68.9	73.4	76.4	78.2	**76.5**	43.5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011–31[1]	57.0	62.0	62.8	66.8	69.1	72.2	**78.3**	80.1
A2. New public sector loans on less favorable terms in 2011–31[2]	57.0	64.7	72.3	79.2	85.0	89.4	**98.4**	70.2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012–13	57.0	65.3	68.9	73.4	76.4	78.2	**76.5**	43.5
B2. Export value growth at historical average minus one standard deviation in 2012–13[3]	57.0	68.5	81.2	86.0	89.0	90.8	**87.3**	47.8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012–13[3]	57.0	65.3	68.9	73.4	76.4	78.2	**76.5**	43.5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012–13[4]	57.0	74.0	88.2	92.7	95.1	96.7	**90.5**	46.7
B5. Combination of B1-B4 using one-half standard deviation shocks	57.0	69.1	81.0	85.3	87.9	89.5	**84.7**	44.8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012[5]	57.0	65.3	68.9	73.4	76.4	78.2	**76.5**	43.5
PV of debt-to-revenue ratio								
Baseline	72.2	76.2	76.6	75.4	75.8	74.0	**62.0**	32.3
A. Alternative Scenarios								
A1. Key variables at their historical average in 2011–31[1]	72.2	72.4	69.9	68.6	68.6	68.4	**63.5**	59.4
A2. New public sector loans on less favorable terms in 2011–31[2]	72.2	75.5	80.4	81.3	84.4	84.7	**79.8**	52.1
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012–13	72.2	78.9	82.4	81.1	81.5	79.6	**66.7**	34.7
B2. Export value growth at historical average minus one standard deviation in 2012–13[3]	72.2	78.3	84.5	82.7	82.6	80.5	**66.2**	33.2
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012–13[3]	72.2	82.6	87.6	86.2	86.6	84.6	**70.9**	36.9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012–13[4]	72.2	86.4	98.2	95.2	94.4	91.6	**73.4**	34.7
B5. Combination of B1-B4 using one-half standard deviation shocks	72.2	89.1	104.5	101.8	101.3	98.4	**79.7**	38.6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012[5]	72.2	105.5	106.1	104.4	104.9	102.5	**85.9**	44.7

Table 2b.Kenya: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011-2031 (continued)
(Percent)

Debt service-to-exports ratio								
Baseline	3.8	4.2	4.2	4.7	4.8	4.9	**5.1**	3.9
A. Alternative Scenarios								
A1. Key variables at their historical average in 2011–31[1]	3.8	4.3	4.3	4.7	4.8	5.0	**5.2**	5.4
A2. New public sector loans on less favorable terms in 2011–31[2]	3.8	4.2	3.9	4.6	4.9	5.3	**6.2**	5.4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012–13	3.8	4.2	4.2	4.7	4.8	4.9	**5.1**	3.9
B2. Export value growth at historical average minus one standard deviation in 2012–13[3]	3.8	4.2	4.5	5.2	5.3	5.5	**6.0**	4.3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012–13[3]	3.8	4.2	4.2	4.7	4.8	4.9	**5.1**	3.9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012–13[4]	3.8	4.2	4.4	5.2	5.3	5.5	**6.4**	4.4
B5. Combination of B1-B4 using one-half standard deviation shocks	3.8	4.1	4.3	4.9	5.0	5.2	**5.9**	4.1
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012[5]	3.8	4.2	4.2	4.7	4.8	4.9	**5.1**	3.9
Debt service-to-revenue ratio								
Baseline	4.8	4.9	4.7	4.8	4.7	4.7	**4.2**	2.9
A. Alternative Scenarios								
A1. Key variables at their historical average in 2011–31[1]	4.8	5.0	4.7	4.8	4.8	4.8	**4.2**	4.0
A2. New public sector loans on less favorable terms in 2011–31[2]	4.8	4.9	4.3	4.8	4.9	5.1	**5.1**	4.0
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012–13	4.8	5.0	5.0	5.1	5.1	5.0	**4.5**	3.1
B2. Export value growth at historical average minus one standard deviation in 2012–13[3]	4.8	4.9	4.7	5.0	4.9	4.9	**4.5**	3.0
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012–13[3]	4.8	5.3	5.3	5.5	5.4	5.3	**4.8**	3.3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012–134	4.8	4.9	4.9	5.3	5.3	5.2	**5.2**	3.2
B5. Combination of B1-B4 using one-half standard deviation shocks	4.8	5.2	5.5	5.9	5.8	5.7	**5.6**	3.6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012[5]	4.8	6.7	6.5	6.6	6.5	6.5	**5.8**	4.0
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline)[6]	17.1	17.1	17.1	17.1	17.1	17.1	**17.1**	17.1

Sources: Country authorities; and staff estimates and projections.

[1] Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

[2] Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

[3] Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

[4] Includes official and private transfers and FDI.

[5] Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

[6] Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.